Explore Image Filtering

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Use the `starter.m` file to complete the following exercises. Be sure to change the placeholder lines of the starter file to match specific information for this assignment.

Download the greyscale image from the website, `Moon.mat`. This image is 800×1200 in size. The code in the starter file will load and display the image.

Important note: The MATLAB command `imagesc` automatically scales the image to use the full dynamic range of the colormap and often produces visually deceiving results for images having maximum pixel values near zero.

1. A point-spread function $\underline{h}_a$ that blurs an image is

$$\underline{h}_a = \frac{1}{36}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

a) The starter file uses the `conv2` command in MATLAB to convolve the image `im` with the point-spread function $\underline{h}_a$ and display the result.. What type of filter does blurring correspond to?

b) Now fill in the `??` part of the starter file to use the `conv2` command in MATLAB to convolve the image `im` with the 16 by 16 point-spread function

$$\underline{h}_b = \frac{1}{256}\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \cdots & 1 \\ & & \vdots & & \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix}$$

and display the result. Discuss what is different from your result in part a).

c) Suppose we first blur the image in the horizontal direction with the point spread function

$$\underline{h}_h = \frac{1}{6}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

and then blur that result in the vertical direction with the point spread function $\underline{h}_v = \underline{h}_h'$, where $\underline{h}_h'$ is the transpose of $\underline{h}_h$. The vectors $\underline{h}_h$ and $\underline{h}_v$ are constructed in the starter file. Will the result of this twice-blurred image be the same as your result from part a)? Why or why not? (Hint: recall the associative property of convolution.) Display the result of the image that was first blurred in the horizontal direction then blurred in the vertical direction.

The starter file assesses the difference between the images using the root mean square difference, `norm(ima-im_hv, 'fro')`. Applying `imagesc` to the difference is deceiving because it automatically scales the input to fit the full visual dynamic range.

2. Another type of point spread function is one that takes a partial derivative in a direction. Point spread function $\underline{h}_{\partial h}$ takes a partial derivative in the horizontal direction and $\underline{h}_{\partial v}$ takes a partial derivative in the vertical direction. The point spread functions for these are given by

$$\underline{h}_{\partial h} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}, \qquad \underline{h}_{\partial v} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}.$$

a) Convolve the point-spread function $\underline{h}_{\partial h}$ with `im` and call this result `imhoriz`. Display the result `imhoriz`.

b) Convolve the point-spread function $\underline{h}_{\partial v}$ with `im` and call this result `imvert`. Display the result `imvert`.

c) Discuss what you see in these images. Note that the pixel intensities in the original 8-bit image are values 0 to 255 with zero representing black and 255 representing white. Forming differences of positive numbers may result in negative values. The command we are using to display the result, `imagesc`, automatically scales the values being displayed to occupy the full grayscale range with the smallest (most negative) values being black and the largest (most positive) values being white.

i) What do white pixels represent in the partial derivative images?

ii) What do black pixels represent in the partial derivative images?

iii) What do grey (neutral) pixels represent in the partial derivative images?

iv) Explain how `imhoriz` and `imvert` are related to the original image `im`?

3. A Laplacian point-spread function approximates a second derivative. Consider the Laplacian point-spread function

$$\underline{L} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

a) Convolve the Laplacian point-spread function $\underline{L}$ with `im` and display the result.

b) What might be one use for a Laplacian point-spread function?